HPE GreenLake for Microsoft Azure Stack HCI

22H2 to 23H2 Version Upgrade Guide

Abstract

This document supports HPE GreenLake for Microsoft Azure Stack HCI release version 1.1.2025.1.

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Document revision history

Document edition	Date	Solution release version
1.0	Jan 2025	1.1.2025.1

High-level workflow of upgrade process

To upgrade the Microsoft Azure Stack HCI from an old version, perform the following high level steps:

1. Upgrade the old operating system (22H2) to the new operating system (23H2) using one of the following methods:
   – Via PowerShell (recommended by Microsoft)
   – Via Windows Admin Center
   – Via other manual methods
2. Perform the post-OS upgrade tasks.
3. Install and enable the Network ATC.
4. Validate the solution upgrade readiness.
5. Apply the solution upgrade.

The following diagram illustrates the Microsoft Azure Stack HCI upgrade process:

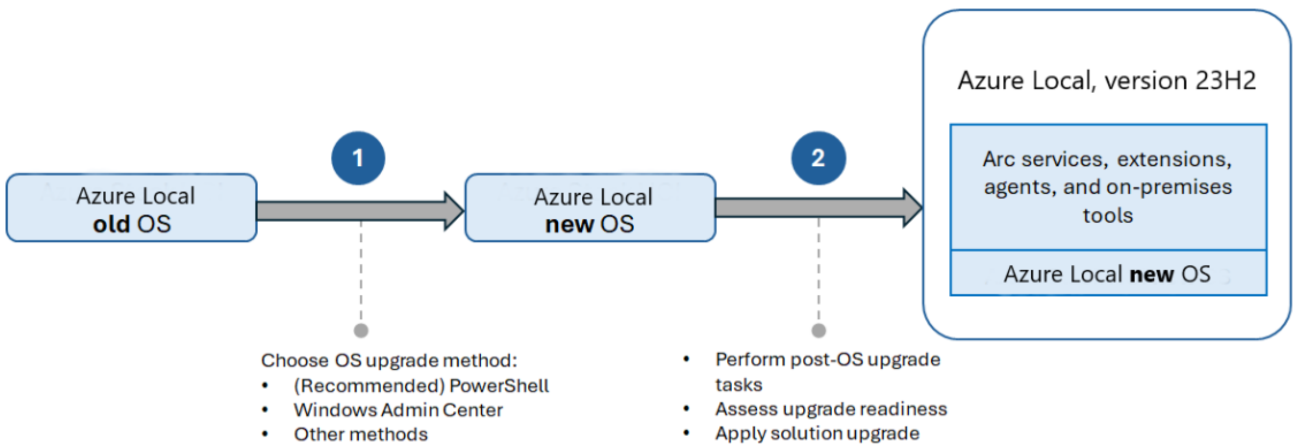


Figure 1. Microsoft Azure Stack HCI upgrade process

Important note /supported services

While proceeding with Microsoft Azure Stack HCI upgrade, note the following pointers about the following services and workloads:

Table 1. Important note/supported services

Workload/Configuration	Currently supported
Azure Kubernetes (AKS) on Microsoft Azure Stack HCI	See notes Kubernetes versions are incompatible between Microsoft Azure Stack HCI, version 22H2, and version 23H2. Remove AKS and all the settings from AKS enabled by Microsoft Azure Arc before you apply the solution upgrade.
Arc VMs on Microsoft Azure Stack HCI	See notes Preview versions of Arc VMs can't be upgraded.
Stretched clusters on Microsoft Azure Stack HCI	No The solution upgrade isn't available for stretched clusters at the time this document is published.

Workload/Configuration	Currently supported
System Center Virtual Machine Manager (SCVMM)	Yes If your Microsoft Azure Stack HCI, version 22H2 cluster is managed by SCVMM 2025, the OS upgrade is supported.
Microsoft Azure Stack HCI, version 22H2SP	No This upgrade process isn't supported for upgrading from Microsoft Azure Stack HCI, version 22H2 Supplemental Package clusters.

High-level workflow for the operating system upgrade

The Microsoft Azure Stack HCI operating system update is available via Windows Update.

To upgrade the operating system on your cluster, follow these high-level steps:

1. Complete the prerequisites.
2. Connect to the Microsoft Azure Stack HCI, version 22H2 cluster.
3. Check for the available updates using Windows Admin Center.
4. Install the new OS, hardware and extension updates using Windows Admin Center.
5. Perform the post-OS upgrade steps.

Complete prerequisites

Check the following points as part of prerequisites to start the Operating System Upgrade Procedure:

1. You have access to an Microsoft Azure Stack HCI, version 22H2 cluster.
2. The cluster should be registered in Microsoft Azure.
3. Make sure that all the nodes in your Microsoft Azure Stack HCI, version 22H2 cluster are **healthy** and show as **Online**.
4. You have access to the Microsoft Azure Stack HCI, version 23H2 OS software update. This update is available via Windows Update. (CAU-Cluster Aware Update).
5. You have access to a client that can connect to your Microsoft Azure Stack HCI cluster. This client should have Windows Admin Center installed on it.

Step 1: Operating system upgrade from 22H2 to 23H2

Step 1: Connect to Microsoft Azure Stack HCI cluster via Windows Admin Center

Perform the following steps to add and connect to a Microsoft Azure Stack HCI server via Windows Admin Center.

1. Select **+ Add** under **All Connections**.
2. Scroll down to Server clusters and select **Add**.
3. Type the name of the cluster and, if prompted, the credentials to use.
4. Select **Add** to finish.
5. The cluster and nodes are added to your connection list on the **Overview** page. Select the cluster to connect to it.

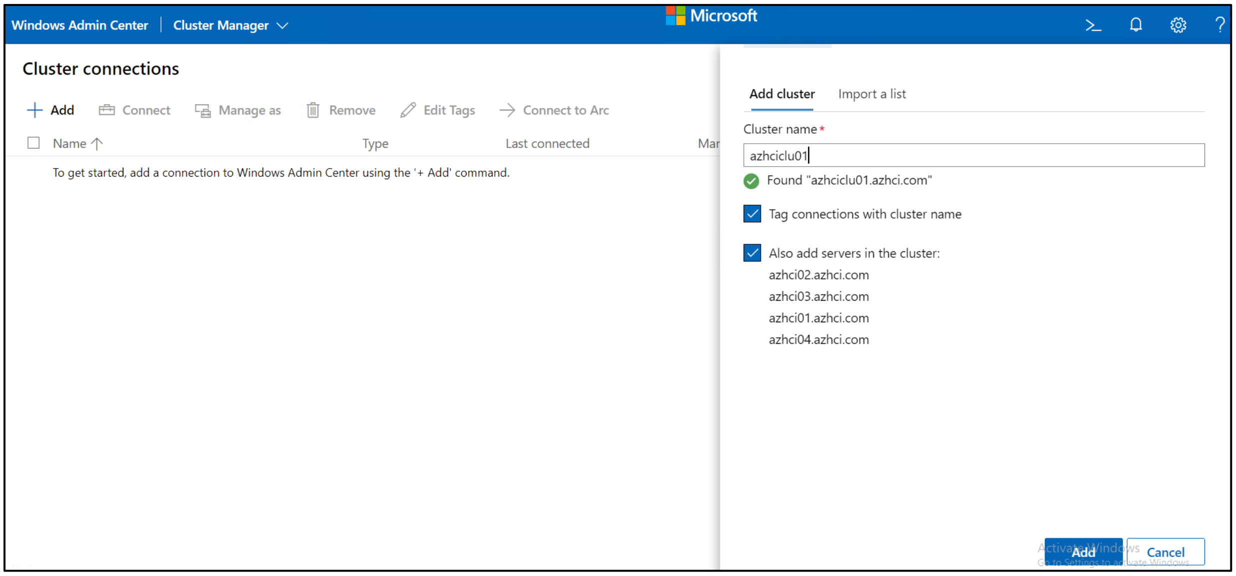


Figure 2. Connect to the cluster

Step 2: Operating system and hardware updates using Windows Admin Center

Windows Admin Center makes it easy to update a cluster and apply quality updates using a simple user interface.

If you purchase an integrated system from a Microsoft hardware partner, it's easy to get the latest drivers, firmware, and other updates directly from Windows Admin Center by installing the appropriate partner update extensions.

If your hardware wasn't purchased as an integrated system, firmware and driver updates would need to be performed separately, following the hardware vendor's recommendations.

Warning

If you begin the update process using Windows Admin Center, continue using the wizard until updates complete. Don't attempt to use the Cluster-Aware Updating (CAU) tool or update a cluster with PowerShell after partially completing the update process in Windows Admin Center. If you wish to use PowerShell to perform the updates instead of Windows Admin Center, see Update a cluster using PowerShell.

Perform the following steps to install updates:

The following steps use Windows Admin Center version 2311. If you are using a different version, your screens may vary slightly.

1. When you connect to a cluster, the Windows Admin Center dashboard alerts you if one or more servers have updates ready to be installed and provide a link to update now. Alternatively, select **Updates** from the **Operations** menu at the left.

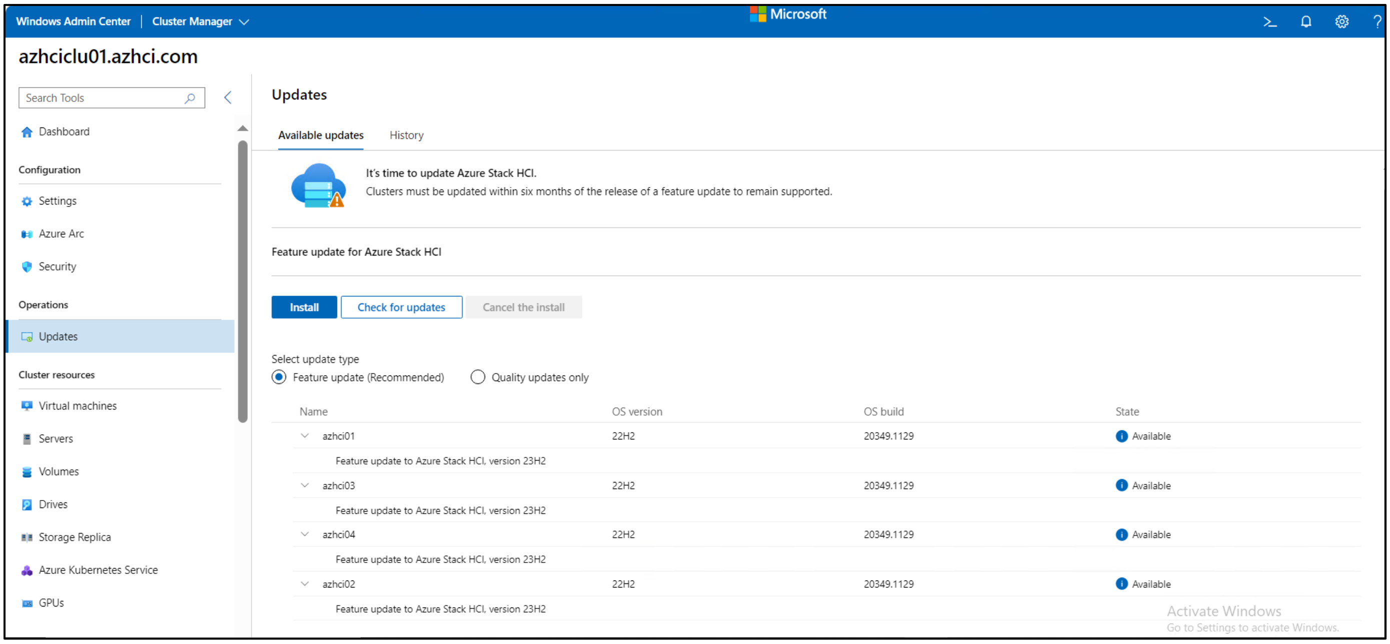


Figure 3. 23H2 feature updates

Note

To use the CAU tool in Windows Admin Center, you must enable Credential Security Service Provider (CredSSP) and provide explicit credentials. If you are asked if CredSSP should be enabled, select **Yes**. Specify your username and password and select **Continue**.

---

2. After the role is installed, Windows Admin Center automatically checks for updates applicable to your cluster.
3. Ensure the radio button for **Feature update (Recommended)** is selected and the **Feature update for Microsoft Azure Stack HCI, version 23H2** is Available for the cluster nodes.
4. If the feature update isn't displayed, ensure your cluster is running the Microsoft Azure Stack HCI OS and that the nodes have direct access to Windows Update, then select **Check for updates**.

Important

Feature updates aren't available in Windows Server Update Services (WSUS).

If you navigate away from the Updates screen while an update is in progress, there could be unexpected behavior, such as the history section of the Updates page not populating correctly until the current run is finished. We recommend opening Windows Admin Center in a new browser tab or window if you wish to continue using the application while the updates are in progress.

5. Select **Install**. Windows Admin Center automatically performs a series of readiness checks to identify issues that could prevent CAU from completing successfully. If any issues are found, select the Details link next to the issue, address the issue, and then select **Check again** to run the readiness checks again.

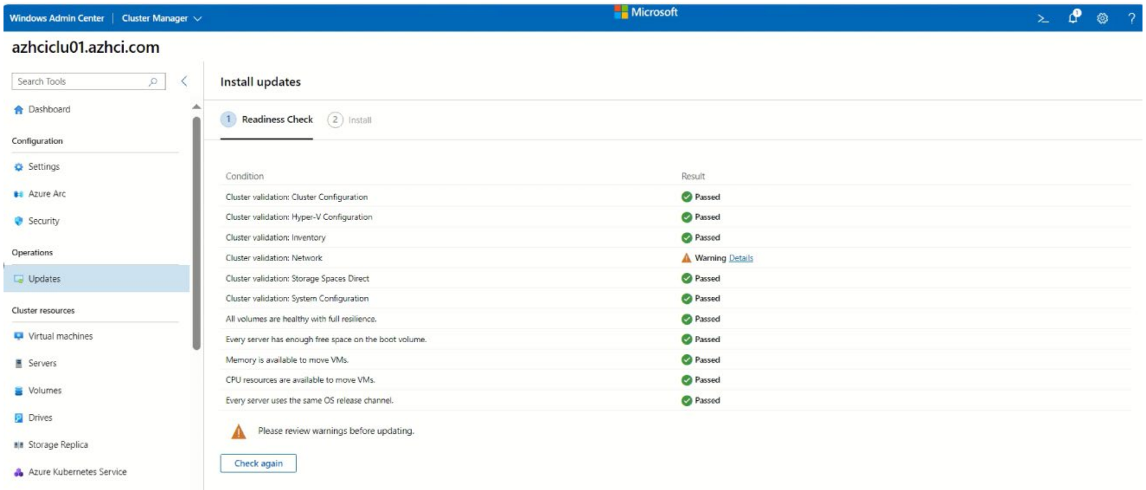
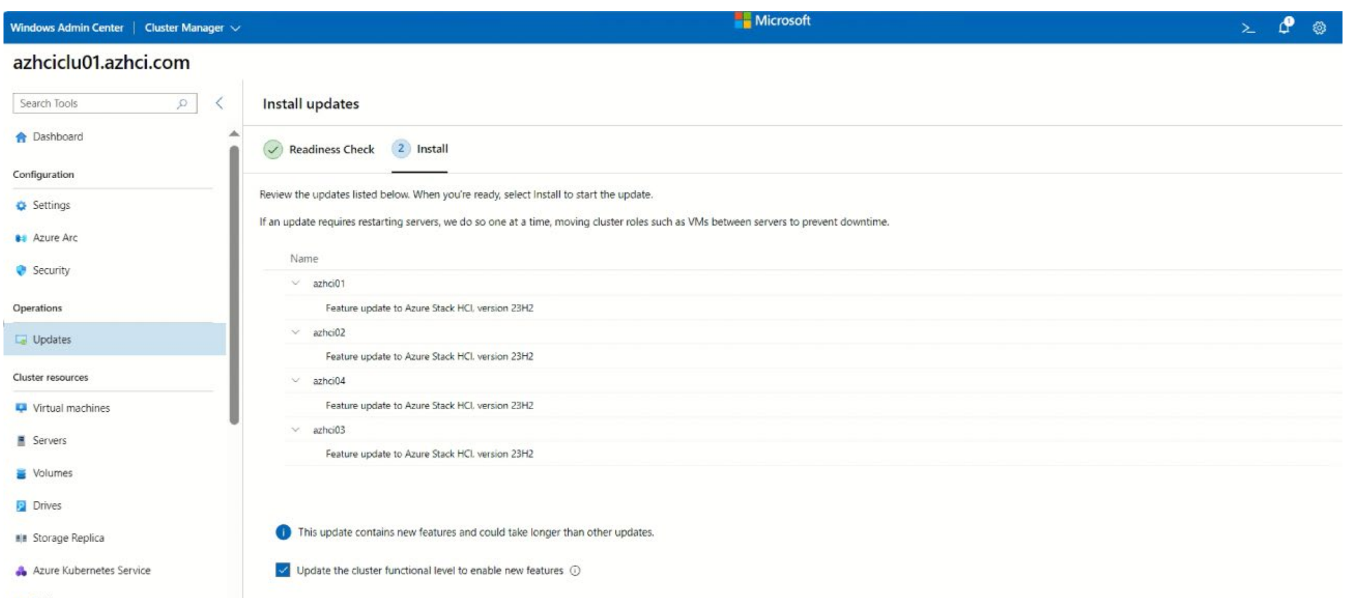


Figure 4. Readiness checks

Note

If you're installing updates on a cluster that has Kernel Soft Reboot enabled, select **Disable Kernel Soft Reboot** for this run checkbox. This selection disables Kernel Soft Reboot as the upgrade requires a full reboot.

5. Select **Next: Install to review** the list of updates to be installed to each cluster node. Then, select Install to begin installing the operating system updates. One by one, each server downloads and applies the updates. The update status changes to Installing updates. If the updates require a restart, servers are restarted one at a time, moving cluster roles such as virtual machines between servers to prevent downtime. Depending on the updates being installed, the entire update run can take anywhere from a few minutes to several hours. You would need to sign in to Windows Admin Center multiple times.



Note

If the updates fail with a **Couldn't install updates** or **Couldn't check for updates** warning or if one or more servers indicate couldn't get status during the run, wait a few minutes, and refresh your browser. You can also use **Get-CauRun** to check the status of the update run with PowerShell.

6. When operating system updates are complete, the update status changes to **Succeeded**. Select **Next: Hardware updates** to proceed to the hardware updates screen.

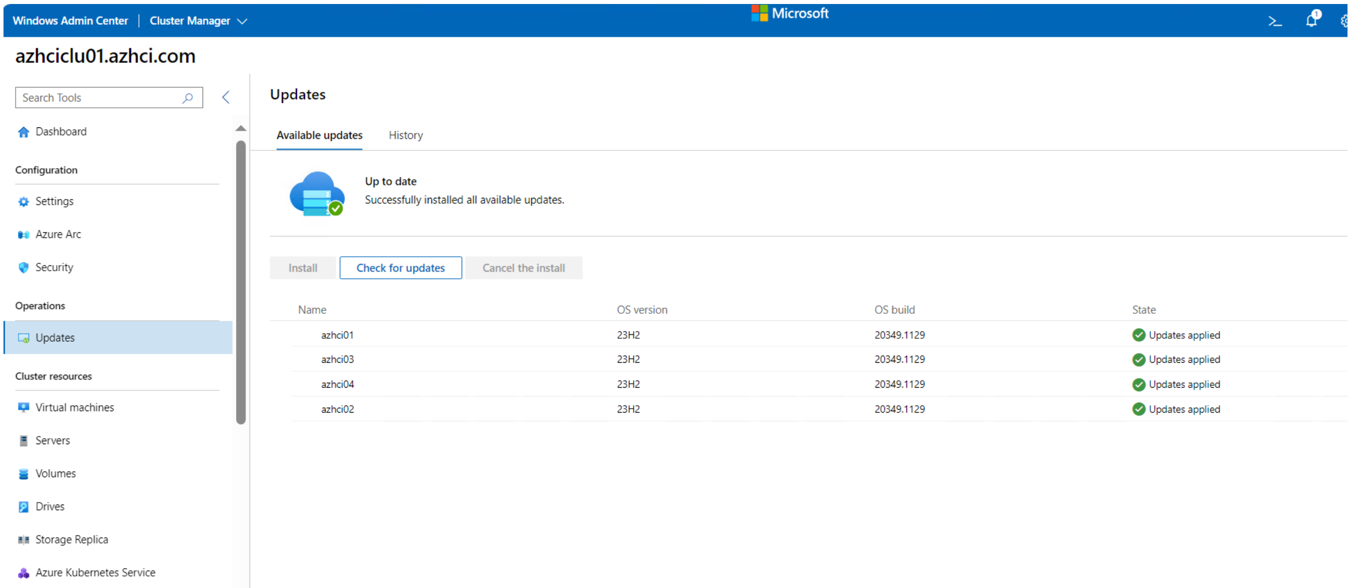


Figure 5. Updates

Important

After applying operating system updates, you may see a message that "storage isn't complete or up-to-date, so we need to sync it with data from other servers in the cluster." This is normal after a server restarts. Don't remove any drives or restart any servers in the cluster until you see a confirmation that the sync is complete.

Note

Hardware updates are only available on clusters that have the vendor's hardware extension installed. If your Windows Admin Center does not have this extension, there will not be an option to install hardware updates.

7. Windows Admin Center checks the cluster for installed extensions that support your specific server hardware. Select **Next: Install to install the hardware updates** on each server in the cluster. If no extensions or updates are found, select **Exit**.

You're now ready to perform the post-upgrade steps for your cluster.

Step 2: Perform post OS-upgrade tasks

After the new operating system is installed, you need to upgrade the cluster functional level and upgrade the storage pool version using PowerShell in order to enable new features.

Update cluster functional level

Upgrade the cluster functional level. (Skip this step if you installed the feature upgrades with Windows Admin Center and checked the optional **Update the cluster functional level to enable new features** checkbox.)

Run the following cmdlet on any server in the cluster: "`Update-ClusterFunctionalLevel`".

Update storage pool version

1. Upgrade the storage pool.

2. After the cluster functional level is upgraded, use the following cmdlet to identify the FriendlyName of the storage pool representing your cluster.

   “Get-StoragePool”

3. Run the Update-StoragePool cmdlet to upgrade the storage pool version.

   “Update-StoragePool -FriendlyName ”S2D on hci-cluster1”

Note

Before you apply the solution upgrade, make sure to install and enable Network ATC on your existing Microsoft Azure Stack HCI cluster. If Network ATC is already enabled on your existing cluster, you can skip this step.

We recommend that you set up Network ATC after you have upgraded the operating system from version 22H2 to version 23H2.

Note

For applying the solution upgrade, it is mandatory to install and enable Network ATC on your existing Microsoft Azure Stack HCI Cluster.

In order to enable and apply Network ATC, it is important to have similar type of Network Adapters in the SET together.

With Current Configuration, we see that OCP and PCIe Card are combined together. This will result in an error while applying Network ATC and creating intents as OCP and PCIe Card with lead to have different component ID’s making them asymmetrical.

Hence, we are proceeding with below set of changes to make our existing Microsoft Azure Stack HCI Cluster compliant with Network ATC Requirement before we make 23H2 Solution upgrade**.**

Re-configuration changes to make Cluster Network ATC Compliant

Step 1: Connect to the Microsoft Azure Stack HCI Nodes using Remote Desktop Connectivity Protocol

Run the following commands till step 4 on all nodes one by one.

Step 2: List out the existing virtual switch on the node 1

Execute the following command to list the existing virtual switch:

```
Get-VMSwitch
```

Step 3: Evict OCP3-P2 adapter from existing Compute switch

Running the following command, to remove the Team member “PCIe Slot 10 Port 2” from the existing Compute SET Switch.

```
Remove-VMSwitchTeamMember -VMSwitchName ”computevswitch” -NetAdapterName ”PCIe Slot 10 Port 2”
```

Step 4: Evict PCIe-P1 adapter from existing mgmt switch

Running the following command, to remove the Team member “PCIe Slot 5 Port 1” from the existing Management SET Switch.

```
Remove-VMSwitchTeamMember -VMSwitchName ”mgmtvswitch” -NetAdapterName ”PCIe Slot 5 Port 1"
```

Step 5: Re-Configuration of Ports in HPE Aruba Switch - 2

Note

Run these commands in only 8325 – HPE Aruba Switch - 2. Do not make any changes on HPE Aruba 8325 switch -1.

---

1. Login into HPE Aruba 8325 – Switch 2. (Considering that PCIe Slot 10 Port 2 and PCIe Slot 5 Port 1) were connected to 8325 Switch 2.
2. Run the following commands to re-configure the ports:

   The following commands depict how to setup a port in trunk mode on switch to allow compute traffic and should be run only on HPE Aruba 8325 switch 2.

```
interface 1/1/1.1-1/1/1.4
 no shutdown
 no routing
 VLAN trunk allowed <<PlatformManagement-vlan>>,<<Compute_vlan>>
```

```
 VLAN trunk native <<Dead_Network_VLAN>>
 exit
write memory
```

The following commands depict how to setup a port in access mode on switch to allow management traffic and should be run only on HPE Aruba 8325 switch 2.

```
interface 1/1/9.1-1/1/9.4
 no shutdown
 no routing
 VLAN access <<PlatformManagement_vlan>>
 exit
write memory
```

Note

Run the following commands on all the Nodes in Microsoft Azure Stack HCI Cluster.

Step 6: Add Team Member back to Compute Switch

Running the following command, we add the Team member "`PCIe Slot 5 Port 1`" to the existing Compute SET Switch.

```
Add-VMSwitchTeamMember -VMSwitchName "computevswitch" -NetAdapterName "PCIe Slot 5 Port 1"
```

Step 7: Add Team Member Back to Mgmt Switch

Running the following command, we add the Team member "`PCIe Slot 10 Port 2`" to the existing Management SET Switch.

```
Add-VMSwitchTeamMember -VMSwitchName "mgmtvswitch" -NetAdapterName "PCIe Slot 10 Port 2"
```

After we have executed the above-mentioned commands on All Cluster Nodes, the configuration has to be as per following table.

Table 2. VLAN and traffic details in 3NIC/6Port configuration VLAN for HPE ProLiant DL380 Gen10 Plus server

Traffic type	VLAN variable	To be defined on switch	Interface on HPE ProLiant DL380 Gen10 Plus server	Access/Trunk
Platform Management	<<PlatformManagement_vlan>>	Aruba 8325 Switch 1 and Aruba 8325 Switch 2	OCP3 - P1, OCP3 - P2	Access
Storage	<<Storage1_vlan>>	Aruba 8325 Switch 1 and Aruba 8325 Switch 2	PCIe2 - P2	Trunk
Storage	<<Storage2_vlan>>	Aruba 8325 Switch 1 and Aruba 8325 Switch 2	PCIe5 - P2	Trunk
Compute	<<Compute_vlan>>	Aruba 8325 Switch 1 and Aruba 8325 Switch 2	PCIe2 - P1, PCIe5 – P1	Trunk

Required Windows roles and features to be installed

Version 23H2 requires a set of Windows roles and features to be installed. Use the following commands for each machine to install the required features. If a feature is already present, the install automatically skips it.

#Install Windows Roles & Features

```
$windowsFeature =  @(
"Failover-Clustering",
                "NetworkATC",
                "RSAT-AD-Powershell",
```

```
                    ”RSAT-Hyper-V-Tools”,
                    ”Data-Center-Bridging”,
                    ”NetworkVirtualization”,
                    ”RSAT-AD-AdminCenter”
                    )
foreach ($feature in $windowsFeature)
{
Install-WindowsFeature -Name $feature -IncludeAllSubFeature -IncludeManagementTools
}

#Install requires optional Windows features
$windowsOptionalFeature = @(

”Server-Core”,
                    ”ServerManager-Core-RSAT”,
                    ”ServerManager-Core-RSAT-Role-Tools”,
                    ”ServerManager-Core-RSAT-Feature-Tools”,
                    ”DataCenterBridging-LLDP-Tools”,
                    ”Microsoft-Hyper-V”,
                    ”Microsoft-Hyper-V-Offline”,
                    ”Microsoft-Hyper-V-Online”,
                    ”RSAT-Hyper-V-Tools-Feature”,
                    ”Microsoft-Hyper-V-Management-PowerShell”,
                    ”NetworkVirtualization”,
                    ”RSAT-AD-Tools-Feature”,
                    ”RSAT-ADDS-Tools-Feature”,
                    ”DirectoryServices-DomainController-Tools”,
                    ”ActiveDirectory-PowerShell”,
                    ”DirectoryServices-AdministrativeCenter”,
                    ”DNS-Server-Tools”,
                    ”EnhancedStorage”,
                    ”WCF-Services45”,
                    ”WCF-TCP-PortSharing45”,
                    ”NetworkController”,
                    ”NetFx4ServerFeatures”,
                    ”NetFx4”,
                    ”MicrosoftWindowsPowerShellRoot”,
                    ”MicrosoftWindowsPowerShell”,
                    ”Server-Psh-Cmdlets”,
                    ”KeyDistributionService-PSH-Cmdlets”,
```

```
                "TlsSessionTicketKey-PSH-Cmdlets",
                "Tpm-PSH-Cmdlets",
                "FSRM-Infrastructure",
                "ServerCore-WOW64",
                "SmbDirect",
                "FailoverCluster-AdminPak",
                "Windows-Defender",
                "SMBBW",
                "FailoverCluster-FullServer",
                "FailoverCluster-PowerShell",
                "Microsoft-Windows-GroupPolicy-ServerAdminTools-Update",
                "DataCenterBridging",
                "BitLocker",
                "Dedup-Core",
                "FileServerVSSAgent",
                "FileAndStorage-Services",
                "Storage-Services",
                "File-Services",
                "CoreFileServer",
                "SystemDataArchiver",
                "ServerCoreFonts-NonCritical-Fonts-MinConsoleFonts",
                "ServerCoreFonts-NonCritical-Fonts-BitmapFonts",
                "ServerCoreFonts-NonCritical-Fonts-TrueType",
                "ServerCoreFonts-NonCritical-Fonts-UAPFonts",
                "ServerCoreFonts-NonCritical-Fonts-Support",
                "ServerCore-Drivers-General",
                "ServerCore-Drivers-General-WOW64",
                "NetworkATC"
            )
foreach ($featureName in $windowsOptionalFeature)
{
Enable-WindowsOptionalFeature -FeatureName $featurename -All -Online
}
```

Step 3: Install and enable Network ATC

Prerequisites

Before installing the Network ATC, take a backup of existing QoS and RDMA configuration, following PowerShell command can be used to get the details:

```
Get-NetAdapterQos | select Name,Enabled
```

```
Get-NetAdapterRdma | select Name,Enabled

Get-NetQosPolicy

Get-NetQosTrafficClass

Get-NetQosFlowControl

Get-NetQosDcbxSetting -InterfaceAlias "<network adapter name>"

Get-NetAdapterAdvancedProperty -Name "<network adapter name>" -RegistryKeyword
*JumboPacket/*NetworkDirectTechnology
```

Step 3.1 Install Network ATC and Stop/disable the Network ATC service on all nodes

```
Install-WindowsFeature -Name NetworkATC

Set-Service -Name NetworkATC -StartupType Disabled

Stop-Service -Name NetworkATC
```

Step 3.2 Pause one node in the cluster, all workloads are moved to other nodes.

Run the following command on Node 1:

Note

Make sure the node that is paused here doesn't have Cluster IP active on it.

```
Suspend-ClusterNode -Drain
```

Step 3.3 Remove the existing VMSwitch configuration on the Paused Node. (Node 1 which is paused in earlier step)

In this step, we eliminate any previous configurations, such as VMSwitch, Data Center Bridging (NetQos) policy for RDMA traffic, and Load Balancing Failover (LBFO), which might interfere with Network ATC's ability to implement the new intent.

Although Network ATC attempts to adopt existing configurations with matching names; including NetQos and other settings, it's easier to remove the current configuration and allow Network ATC to redeploy the necessary configuration items and more.

If you have more than one VMSwitch on your system, make sure you specify the switch attached to the adapters being used in the intent.

To remove the existing **Compute VMSwitch** configuration, run the following command:

```
Get-VMSwitch -Name "computevswitch" | Remove-VMSwitch -force
```

To remove the existing **Management VMSwitch** configuration, run the following command:

```
Get-VMSwitch -Name "mgmtvswitch" | Remove-VMSwitch -force
```

Note

Generally, removing mgmt. Switch using the above command will affect any existing RDP connection with the Microsoft Azure Stack HCI Node. Re-connect back to the Node using IP address/Hostname.

Step 3.4 Remove your existing NetQos configuration on the Paused Node Only

Execute the following commands:

```
Get-NetQosTrafficClass | Remove-NetQosTrafficClass

Get-NetQosPolicy | Remove-NetQosPolicy -Confirm:$false

Get-NetQosFlowControl | Disable-NetQosFlowControl
```

Step 3.5 Start the Network ATC service, on the paused node only

As a precaution, to control the speed of the rollout, we paused the node and then stopped and disabled the Network ATC service in the previous steps.

Since Network ATC intents are implemented cluster-wide, perform this step only once.

To start the Network ATC service, on the paused node only, run the following command:

```
Set-service -Name NetworkATC -StartupType Automatic
```

```
Start-Service -Name NetworkATC
```

Step 3.6 Add the Network ATC intent

1. Run the following commands one by one to add the Intent for Fully disaggregated host networking. There are three intents that are managed across cluster nodes in Fully Disaggregated Intent Type (Management, Compute and Storage intents)
2. Please run the following override command to disable rdma and QOS on management and compute intent.

```
$MgmtAdapterPropertyOverrides = New-NetIntentAdapterPropertyOverrides
$MgmtAdapterPropertyOverrides.NetworkDirect = 0
$MgmtAdapterPropertyOverrides.QOS = 0
```

3. To Add Management Intent run the following command: `Add-NetIntent -Name Mgmt -Management -AdapterName "PCIe Slot 10 Port 1","PCIe Slot 10 Port 2" -AdapterPropertyOverrides $MgmtAdapterPropertyOverrides`
4. To Add Compute Intent run the following command: `Add-NetIntent -Name Compute -Compute -AdapterName "PCIe Slot 2 Port 1", "PCIe Slot 5 Port 1" -AdapterPropertyOverrides $MgmtAdapterPropertyOverrides`
5. Before executing the Storage intent Creation command make sure to run the below commands for override Automatic Storage IP addressing as we have our customer VLAN ID's and no requirement for Automatic IP addressing.

```
$StorageOverride = New-NetIntentStorageOverrides
$StorageOverride.EnableAutomaticIPGeneration = $false
$StorAdapterPropertyOverrides = New-NetIntentAdapterPropertyOverrides
$StorAdapterPropertyOverrides.JumboPacket = 9014
```

6. To add Storage Intent run the following command:

```
Add-NetIntent -Name Storage -Storage -AdapterName "PCIe Slot 2 Port 2", "PCIe Slot 5 Port 2" -storagevlans
1703,1704 -StorageOverrides $StorageOverride -AdapterPropertyOverrides $StorAdapterPropertyOverrides
```

Step 3.7 Verify the deployment on one node

To verify your node's successful deployment of the intents submitted in earlier step , run the following command:

```
Get-NetIntentStatus -Name <IntentName>
```

Ensure that each intent added has an entry for the host you're working on.

Note

After creating the Intent we need to go to the failover cluster manager tool and check for live migration setting under **Networks** and make sure necessary networks (storage network1, storage network2 and management) is checked, if the networks is not checked then live migration will fail.

Make sure to execute the following steps only once **Get-NetIntentStatus** and the **ConfigurationStatus** shows **Success** and **ProvisioningStatus** shows **Completed** for all the deployed Intents.

Step 3.8 Rename the VMSwitch on all the other nodes

Note

Renaming the virtual switch is a non-disruptive change and can be done on all the nodes simultaneously.

Run the following commands:

#Run on the node where Network ATC is configured. (Node 1) (This command returns the VMSwitch Name as per Network ATC)

```
Get-VMSwitch | ft Name
```

Example output:

```
PS C:\Users\Administrator.AZHCI> Get-VMSwitch | ft Name

Name
----
ConvergedSwitch(mgmt)
ComputeSwitch(compute)
```

Figure 6. Get-VMSwitch | ft Name

#Run on the next node to rename the virtual switch (Node 2, Node 3, …Node N) (This helps us rename VMSwitch as per Network ATC Names on all the other Nodes).

```
Rename-VMSwitch -Name 'ExistingName' -NewName 'NewATCName'
```

Example output:

```
PS C:\Users\Administrator.AZHCI> Get-VMSwitch

Name           SwitchType NetAdapterInterfaceDescription
----           ---------- ------------------------------
mgmtvswitch    External   Teamed-Interface
computevswitch External   Teamed-Interface


PS C:\Users\Administrator.AZHCI> Get-VMNetworkAdapter -ManagementOS

Name     IsManagementOs VMName SwitchName  MacAddress   Status IPAddresses
----     -------------- ------ ----------  ----------   ------ -----------
mgmtvNIC True                  mgmtvswitch 88E9A44B9CEA {Ok}


PS C:\Users\Administrator.AZHCI> Rename-VMSwitch -Name 'mgmtvswitch' -NewName 'ConvergedSwitch(mgmt)'
PS C:\Users\Administrator.AZHCI> Rename-VMSwitch -Name 'computevswitch' -NewName 'ComputeSwitch(compute)'
PS C:\Users\Administrator.AZHCI> Get-VMSwitch

Name                   SwitchType NetAdapterInterfaceDescription
----                   ---------- ------------------------------
ConvergedSwitch(mgmt)  External   Teamed-Interface
ComputeSwitch(compute) External   Teamed-Interface
```

Figure 7. Rename-VMSwitch -Name 'ExistingName' -NewName 'NewATCName'

Step 3.9 Get the vmnetwork adapter name

On the paused node where network ATC is applied run this command to get the vmnetwork adapter name

```
Get-VMNetworkAdapter -ManagementOS
```

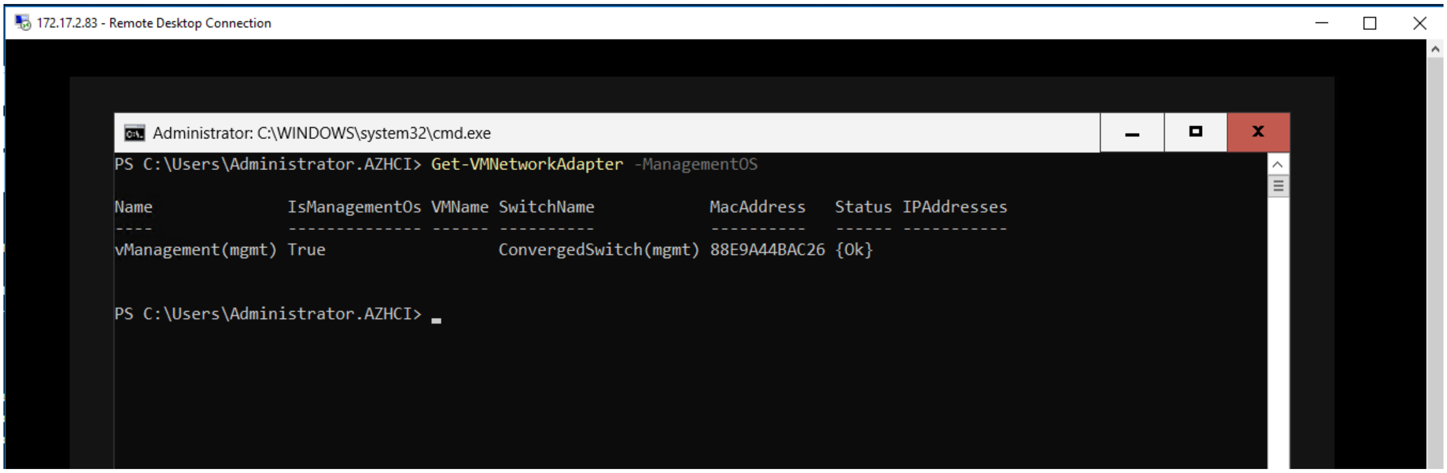


Figure 8. Get-VMNetworkAdapter -ManagementOS

Use the output from the above command and run the following command on all other nodes

```
Rename-vmnetworkadapter -managementOS -name "mgmtvNIC" -newname "vManagement(mgmt)"
```

Run the following command on all other nodes

```
Get-NetAdapter | Where-Object {$_.Name -IMatch "vEthernet"} | Rename-NetAdapter -NewName (Get-VMNetworkAdapter -
ManagementOS).Name
```

Step 3.10: Disconnect and Re-connect VM's with the new updated switch Name

For this activity, there is a downtime that is expected.

Note

(After the virtual switch is renamed, you must disconnect and reconnect each virtual machine so that it can appropriately cache the new name of the virtual switch. This is a disruptive action that requires planning and downtime to complete. If you do not perform this action, live migrations will fail with an error indicating the virtual switch doesn't exist on the destination.)

After your switch is renamed, disconnect and reconnect your vNICs for the VMSwitch name change to go through.

The following command can be used to perform this action for all VMs:

```
$VMSW =  Get-VMSwitch -Name "ComputeSwitch(compute)"
```

```
$VMs = Get-VM
```

```
$VMs | %{Get-VMNetworkAdapter -VMName $_.name | Disconnect-VMNetworkAdapter ; Get-VMNetworkAdapter -VMName
$_.name | Connect-VMNetworkAdapter -SwitchName $VMSW.name}
```

We have got 1ms of downtime as we had only 3 VMs running on that node.

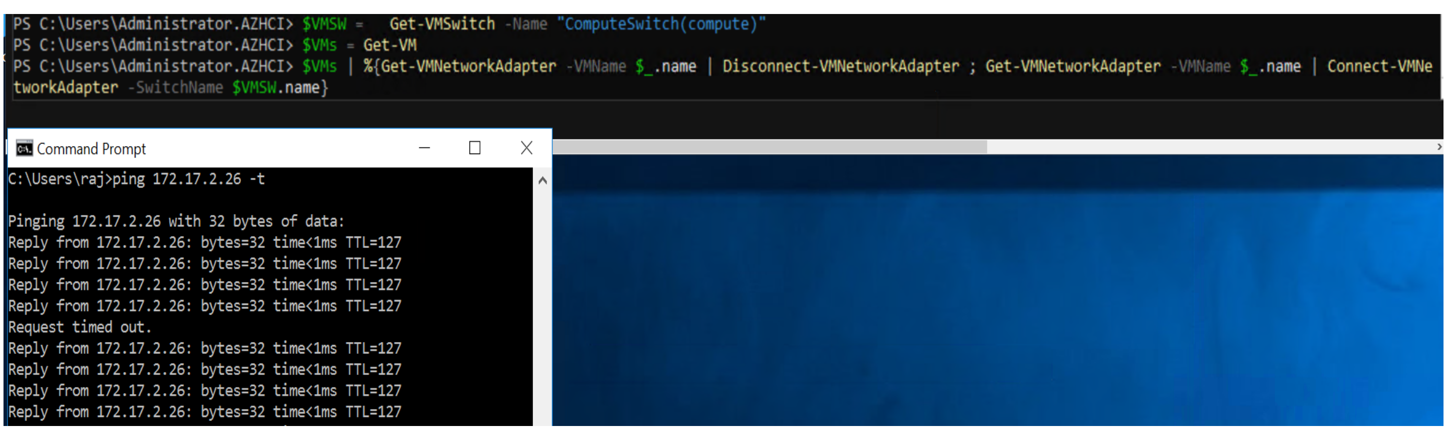


Figure 9. VMSW = Get-VMSwitch -Name "ComputeSwitch(compute)"

Step 3.11 Resume the cluster node

To re-enter or put your cluster back in service, run the following command on the node which was paused earlier: Resume-ClusterNode

Step 3.12 Apply Network ATC across the cluster

To apply the Network ATC settings across the cluster, repeat the following steps on each node one at a time:

1. Repeat Step 3.2 only.
2. Repeat only Step 3.4 and 3.5 (Do not run Step 3.3 as Switches are already renamed on all Nodes.)
3. Repeat only Step 3.7 to verify deployment. Make sure to follow below steps only once **Get-NetIntentStatus** show the **ConfigurationStatus** shows **Success** and **ProvisioningStatus** shows **Completed** for all the deployed Intents.
4. Repeat Step 3.11 to add resume the Paused Node.

Note

The `Get-NetIntentStatus` command shows the deployment status of the requested intents. The result returns one object per intent for each node in the cluster. For example, if you have a three-node cluster with two intents, you should see six objects, each with their own status, returned by the command.

Step 3.13 Post Network ATC observations and changes

1. Jumbopacket value for storage adapter fall back to Default. We have used override to set it to 9014 at step 3.6.
2. SMBDirect is renamed to SMB_Direct after implementing Network ATC. Hence, we need to carefully give the name while validating DCB

   Before(22H2)

```
PS C:\Users\administrator.AZHCI> Get-NetQosTrafficClass

Name                       Algorithm Bandwidth(%) Priority                  PolicySet       IfIndex IfAlias
----                       --------- ------------ --------                  ---------       ------- -------
[Default]                  ETS       49           0-2,4-6                   Global
Cluster                    ETS       1            7                         Global
SMBDirect                  ETS       50           3                         Global


PS C:\Users\administrator.AZHCI>
```

   After(23H2)

```
PS C:\Test> Get-NetQosTrafficClass

Name                       Algorithm Bandwidth(%) Priority                  PolicySet       IfIndex IfAlias
----                       --------- ------------ --------                  ---------       ------- -------
[Default]                  ETS       49           0-2,4-6                   Global
SMB_Direct                 ETS       50           3                         Global
Cluster                    ETS       1            7                         Global
```

3. While running Validate-DCB tool we get error for Live Migration limit, to set Limit of bandwidth for Live Migration, perform the following steps:

   Step 1. Run the following overrides command on any one node

```
$clusterOverride = New-NetIntentGlobalClusterOverrides
$clusterOverride.EnableLiveMigrationNetworkSelection = $true
$clusterOverride.MaximumSMBMigrationBandwidthInGbps = 6
Set-NetIntent -GlobalClusterOverrides $clusterOverride
```

   Check the status for globaloverrides using following command:

```
Get-NetIntentStatus -Globaloverrides
```

```
Select Administrator: C:\WINDOWS\system32\cmd.exe
PS C:\Users\Administrator.AZHCI> Get-NetIntentStatus -Globaloverrides


LastConfigApplied   : 18
LastSuccess         : 01/16/2025 07:17:57
LastUpdated         : 01/16/2025 07:17:57
Progress            : 18 of 18
Error               :
Host                : azhci01
ConfigurationStatus : Success
RetryCount          : 0
ProvisioningStatus  : Completed


LastConfigApplied   : 18
LastSuccess         : 01/16/2025 07:18:12
LastUpdated         : 01/16/2025 07:18:12
Progress            : 18 of 18
Error               :
Host                : azhci02
ConfigurationStatus : Success
RetryCount          : 0
ProvisioningStatus  : Completed
```

Step 2. We must modify the test expectations, (Edit the NIC capability from 10Gbps to 25 Gbps), Open the following file in the server where you're running Validate DCB

`C:\Program Files\WindowsPowerShell\Modules\Validate-DCB\20210802.2.2.117\tests\unit\modal.unit.tests.ps1`

Find line 904:

Ensure values are like this

```
line 892              : {$_ -le 25000000000}
Line 893              : [[[[$thisPolicy.BandwidthPercentage / 100] * .5]
Line 901              : {$_ -gt 25000000000}


Line 902              : $expectedLimitMB = [[[[$thisPolicy.BandwidthPercentage / 100] * .5]
```

```
{$_ -le 25000000000} {
    $expectedLimitMB = (((($thisPolicy.BandwidthPercentage / 100) * .5) * $AdapterLinkSpeed) / 8) / 1000000
    It "Should have a Live Migration limit of less than $expectedLimitMB MBps" {
        $SMBBandwidthLimit.BytesPerSecond / 1MB | Should BeLessThan ($expectedLimitMB + 1)
    }
}

# Setting to a Max of 750 MBps for adapters over 10 Gbps
# https://techcommunity.microsoft.com/t5/failover-clustering/optimizing-hyper-v-live-migrations-on-an-hyperconverged/ba-p/396609
{$_ -gt 25000000000} {
    $expectedLimitMB = (((($thisPolicy.BandwidthPercentage / 100) * .5) * $AdapterLinkSpeed) / 8) / 1000000
    It "Should have an Live Migration limit of 750 MBps" {
        $SMBBandwidthLimit.BytesPerSecond / 1MB | Should Be 750
```

After completion, save the file and close it.

4. Now run Validate-DCB. (Make sure to give name while running as SMB_Direct)

   `Get-NetAdapterRdma | Select Name, Enabled` (We have added AdapterPropertyOverrides at step 3.6 for mgmt. and storage intent)

   Before (22H2):

```
PS C:\Users\administrator.AZHCI> Get-NetAdapterRdma | Select Name,Enabled

Name                    Enabled
----                    -------
PCIe Slot 5 Port 2         True
vEthernet (mgmtvNIC)      False
PCIe Slot 10 Port 1       False
PCIe Slot 2 Port 1        False
PCIe Slot 2 Port 2         True
PCIe Slot 10 Port 2       False
PCIe Slot 5 Port 1        False
```

After (23H2):

```
PS C:\Users\Administrator.AZHCI> Get-NetAdapterRdma | Select Name,Enabled

Name                Enabled
----                -------
PCIe Slot 10 Port 1    True
PCIe Slot 2 Port 1     True
PCIe Slot 5 Port 2     True
PCIe Slot 2 Port 2     True
PCIe Slot 10 Port 2    True
vManagement(mgmt)     False
PCIe Slot 5 Port 1     True
```

5. Get-NetAdapterQos | select Name,Enabled (We have added AdapterPropertyOverrides at step 3.6 for mgmt. and storage intent)

Before (22H2):

```
PS C:\Users\administrator.AZHCI> Get-NetAdapterQos | select Name,Enabled

Name                Enabled
----                -------
PCIe Slot 5 Port 2     True
PCIe Slot 10 Port 1   False
PCIe Slot 2 Port 1    False
PCIe Slot 2 Port 2     True
PCIe Slot 10 Port 2   False
PCIe Slot 5 Port 1    False
```

After (23H2):

```
PS C:\Users\Administrator.AZHCI> Get-NetAdapterQos | select Name,Enabled

Name                Enabled
----                -------
PCIe Slot 10 Port 1    True
PCIe Slot 2 Port 1     True
PCIe Slot 5 Port 2     True
PCIe Slot 2 Port 2     True
PCIe Slot 10 Port 2    True
PCIe Slot 5 Port 1     True
```

6. `Get-NetQosFlowControl`

Before (22H2):

```
PS C:\Users\administrator.AZHCI> Get-NetQosFlowControl

Priority   Enabled    PolicySet        IfIndex IfAlias
--------   -------    ---------        ------- -------
0          False      Global
1          False      Global
2          False      Global
3          True       Global
4          False      Global
5          False      Global
6          False      Global
7          True       Global
```

After (23H2):

```
PS C:\Users\Administrator.AZHCI> Get-NetQosFlowControl

Priority   Enabled    PolicySet        IfIndex IfAlias
--------   -------    ---------        ------- -------
0          False      Global
1          False      Global
2          False      Global
3          True       Global
4          False      Global
5          False      Global
6          False      Global
7          False      Global


PS C:\Users\Administrator.AZHCI>
```

7. Inbox drivers are not supported for use with Azure Local. To identify if your adapter is using an inbox driver, run the following cmdlet. An adapter is using an inbox driver if the **DriverProvider** property is **Microsoft.**

```
Get-NetAdapter -Name <AdapterName> | Select *Driver*
```

```
PS C:\Users\Administrator.AZHCI> Get-NetAdapter -Name "PCIe Slot 5 Port 2" | Select *Driver*


DriverInformation       : Driver Date 2024-04-16 Version 24.4.26429.0 NDIS 6.85
DriverFileName          : mlx5.sys
DriverVersion           : 24.4.26429.0
DriverDate              : 2024-04-16
DriverDateData          : 133576992000000000
DriverDescription       : Mellanox ConnectX-5 Adapter
DriverMajorNdisVersion  : 6
DriverMinorNdisVersion  : 85
DriverName              : \SystemRoot\System32\drivers\mlx5.sys
DriverProvider          : Mellanox Technologies Ltd.
DriverVersionString     : 24.4.26429.0
MajorDriverVersion      : 2
MinorDriverVersion      : 0
```

Step 4: Validate solution upgrade readiness for Microsoft Azure Stack HCI

Assess solution upgrade readiness

The following steps help assess the readiness of your Microsoft Azure Stack HCI cluster for the solution upgrade.

1. Install and use the Environment Checker to verify that Network ATC is installed and enabled on the node. Verify that there are no Preview versions for Arc Resource Bridge running on your cluster.
2. Ensure that sufficient storage space is available for infrastructure volume.
3. Perform other checks such as installation of required and optional Windows features, enablement of Application Control policies, BitLocker suspension, and OS language.
4. Review and remediate the validation checks that block the upgrade.

Use Environment Checker to validate upgrade readiness

Microsoft recommends that the Environment Checker is used to validate system readiness before we upgrade the solution.

For more information, refer Assess environment readiness with Environment Checker.

A report is generated with potential findings that require corrective actions to be ready for the solution update.

Some of the actions require server reboots. The information from the validation report allows you to plan maintenance windows ahead of time to be ready. The same checks are executed during the solution upgrade to ensure your system meets the requirements.

Blocking validation tests for upgrade

The following table contains the validation tests with severity *Critical* that block the upgrade. Any items that block the upgrade must be addressed before you apply the solution upgrade.

Table 1. Blocking validation tests for upgrade

Name	Severity
Windows OS is 23H2	Critical
AKS HCI install state	Critical
Supported cloud type	Critical
BitLocker suspension	Critical
Cluster exists	Critical
All nodes in same cluster	Critical
Cluster node is up	Critical
Stretched cluster	Critical
Language is English	Critical
Microsoft on-premises cloud (MOC) install state	Critical
MOC services running	Critical
Network ATC feature is installed	Critical
Required Windows features	Critical
Storage pool	Critical
Storage volume	Critical
Windows Defender for Application Control (WDAC) enablement	Critical

Table 2. Non-blocking validation tests for upgrade

The following table contains the validation tests with severity *Warning* that should be addressed after the upgrade to take advantage of the new capabilities introduced with Microsoft Azure Stack HCI, version 23H2.

Name	Severity
Trusted Platform Module (TPM) Property OwnerClearDisabled is False	Warning
TPM Property TpmReady is True	Warning
TPM Property TpmPresent is True	Warning
TPM Property LockoutCount is 0	Warning
TPM Property TpmActivated is True	Warning
TPM Property ManagedAuthLevel is Full	Warning
TPM Property AutoProvisioning is Enabled	Warning

Name	Severity
TPM Property LockedOut is False	Warning
TPM Property TpmEnabled is True	Warning

Step 4.1 Set up the Environment Checker

Perform the following steps to set up the Environment Checker on a server node of your Microsoft Azure Stack HCI cluster:

1. Select one server that's the part of the cluster.
2. Sign in to the server using local administrative credentials.
3. Install the Environment Checker on the server. Run the following PowerShell command from the PSGallery:

```
Install-Module -Name AzStackHci.EnvironmentChecker -AllowClobber
```

Step 4.2 Run the validation

1. Sign in to the server where you installed the Environment Checker using local administrative credentials.
2. To run the validation locally on the server, run the following PowerShell command:

```
Invoke-AzStackHciUpgradeValidation
```

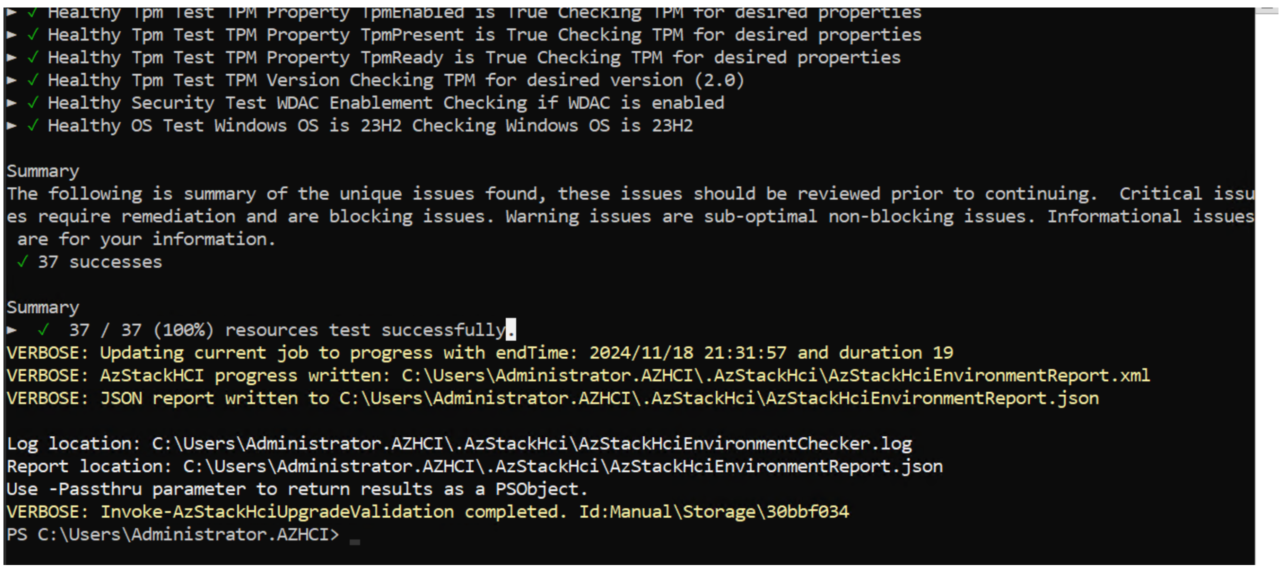


Figure 10. Invoke-AzStackHciUpgradeValidation

Step 5: Apply the solution upgrade

Install solution upgrade on your Microsoft Azure Stack HCI cluster

High Level workflow for solution upgrade

The following is the path for successful deployment:

Environment Checker > Active Directory Preparation > Have Block of 6-Continuous IP > Meet All prerequisites > Deploy from Azure Portal > manage the Deployment > Successful Deployment.

Step 5.1 Prerequisites

1. Validate the system using the Environment Checker as per the instructions mentioned in the section **Validate Solution Upgrade Readiness**.
2. Verify that latest AzureEdgeLifecycleManager extension on each machine is installed on each node in Azure Portal. Navigate to **Azure Portal > Azure Local > All Clusters >Select the Required Cluster > Settings > Extensions**.
3. Prepare Active Directory as per 23H2 Deployment instructions. Make sure that the **Deployment user** created as part of this Active Directory preparation is a **member of the local Administrator group** on all the Nodes.
4. Have IPv4 network range with six, contiguous IP addresses available for new Azure Arc services. Ensure that the IP addresses aren't in use and meet the outbound connectivity requirement.
5. Have Azure subscription permissions for Microsoft Azure Stack HCI Administrator and Reader.

Step 5.2 Active Directory preparation

For Active Directory requirements, refer to the URL as per Microsoft: Prepare Active Directory for Azure Local, version 23H2 deployment - Azure Local | Microsoft Learn.

Step 5.3 Install the solution upgrade via Microsoft Azure portal

1. Make sure to add the "**Deployment User**" created as part of Active Directory prerequisites is added to Local administrator on all nodes in the Cluster.
2. Go to your Azure Local resource in Microsoft Azure portal.
3. In the **Overview** page, click banner indicates that a solution upgrade is available. Select the **Upgrade** link in the banner.
4. On the **Basics** tab, specify the following information:
5. Create a new key vault to store the credentials.
6. Select **Create a new key vault**.
7. Provide a **Name** for the new key vault. The name should be 3 to 24 characters long and contain only letters, numbers, and hyphens. Two consecutive hyphens are not allowed.
8. Click **Grant Key Vault Permissions**.
9. Specify the deployment account credential. This credential is from your Active Directory for a principal that is a member of the local Administrator group on each machine.

Note

The user can't be Administrator and can't use format domain\username.

Deployment user must be added to local administrator on all nodes in cluster.

10. Accept default name for custom location name used for Azure Arc services.
11. Specify network IP address information. A total of six, contiguous IP addresses is required, defined by an IP address range. IP addresses in the range must not be in use, meet outbound connectivity requirements, Communicate to the host IP addresses.

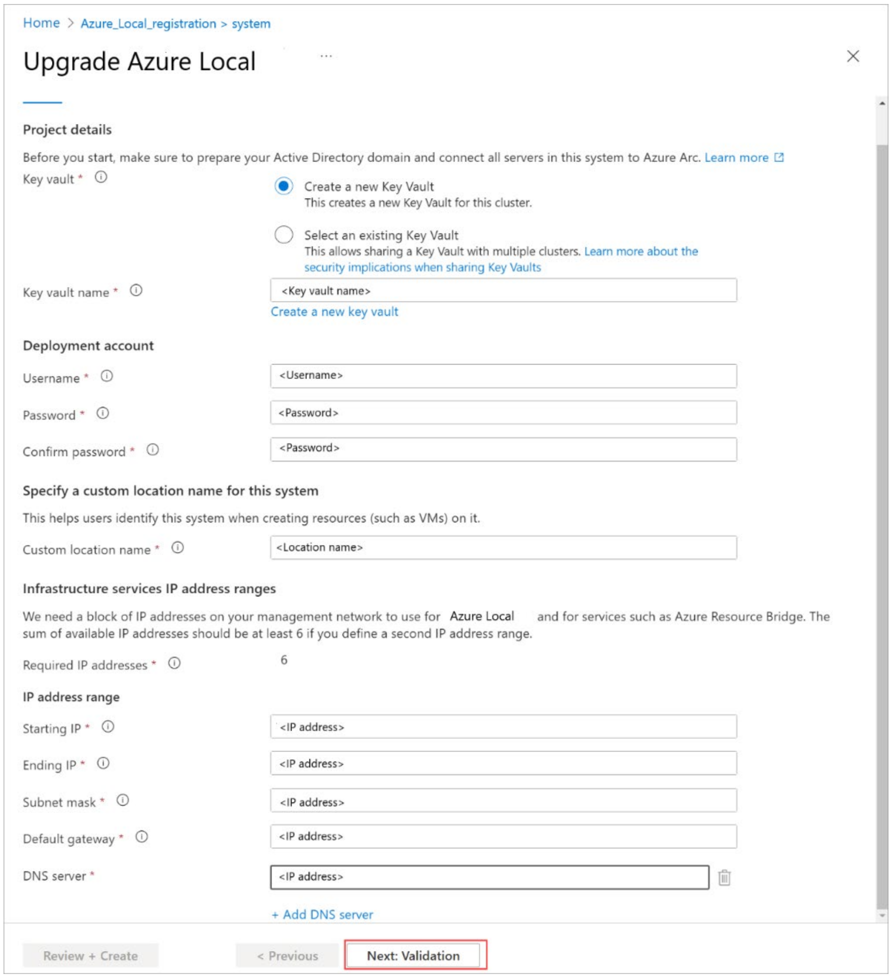


Figure 11. Upgrade Azure Local

12. Select **Next: Validation**.

13. On the **Validation** tab, the operation automatically creates Microsoft Azure resources and also configures permissions and the audit login.

14. Select **Start validation** to begin the operation. This operation involves running the Environment Checker to check external connectivity and storage requirements and that the environment is ready for solution upgrade.

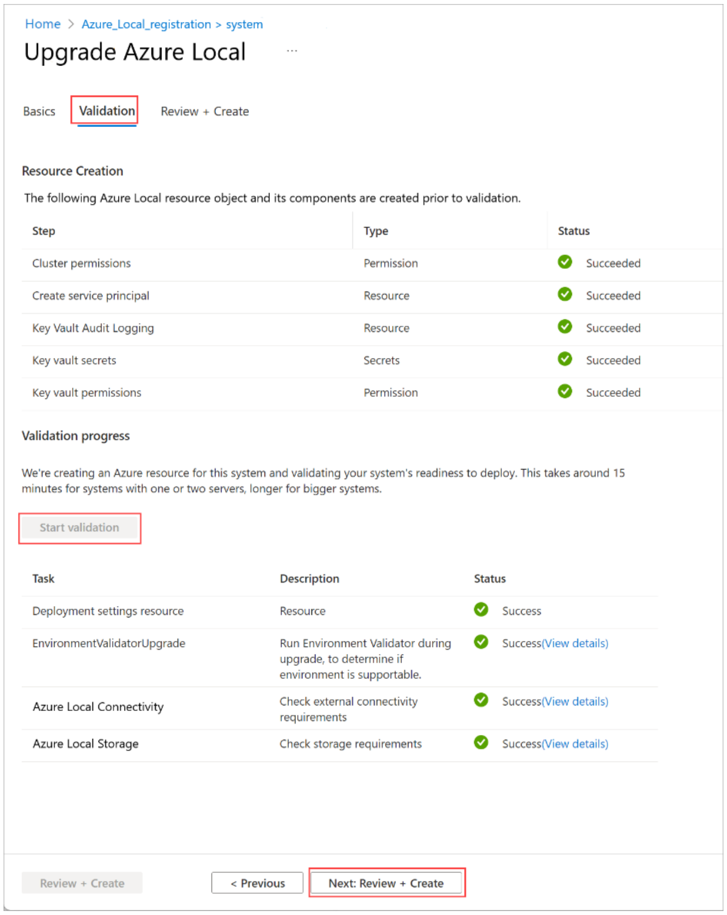


Figure 12. Validation

15. After the validation is complete, select **Next: Review + Create**.

16. On the **Review + Create** tab, review the summary for the solution upgrade.

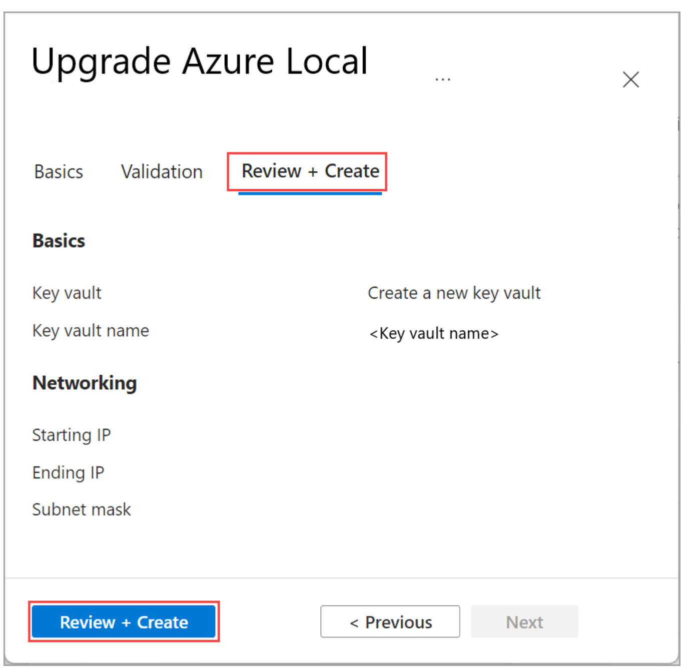


Figure 13. Review + Create

17. Select **Review + Create** to start the upgrade process. A notification is displayed that the deployment is in progress.

Step 5.4 Monitor the Deployment Process via Microsoft Azure portal

1. After the upgrade starts, you are automatically taken to **Settings > Deployment**. Refresh the screen periodically and monitor the upgrade progress.

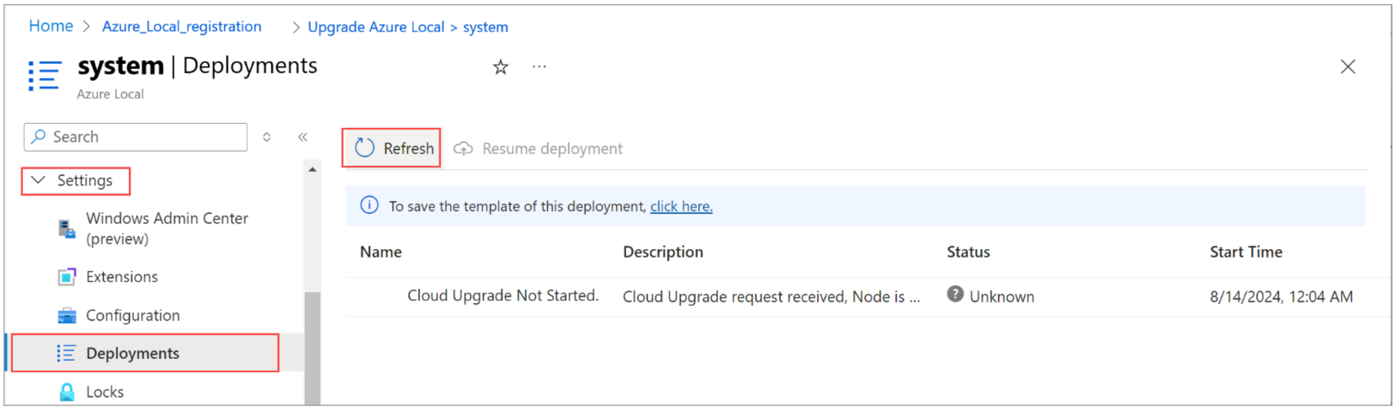


Figure 14. Settings - Deployments

2. Wait for the upgrade to complete. The solution upgrade process can take a few hours depending upon the number of machines in the system.

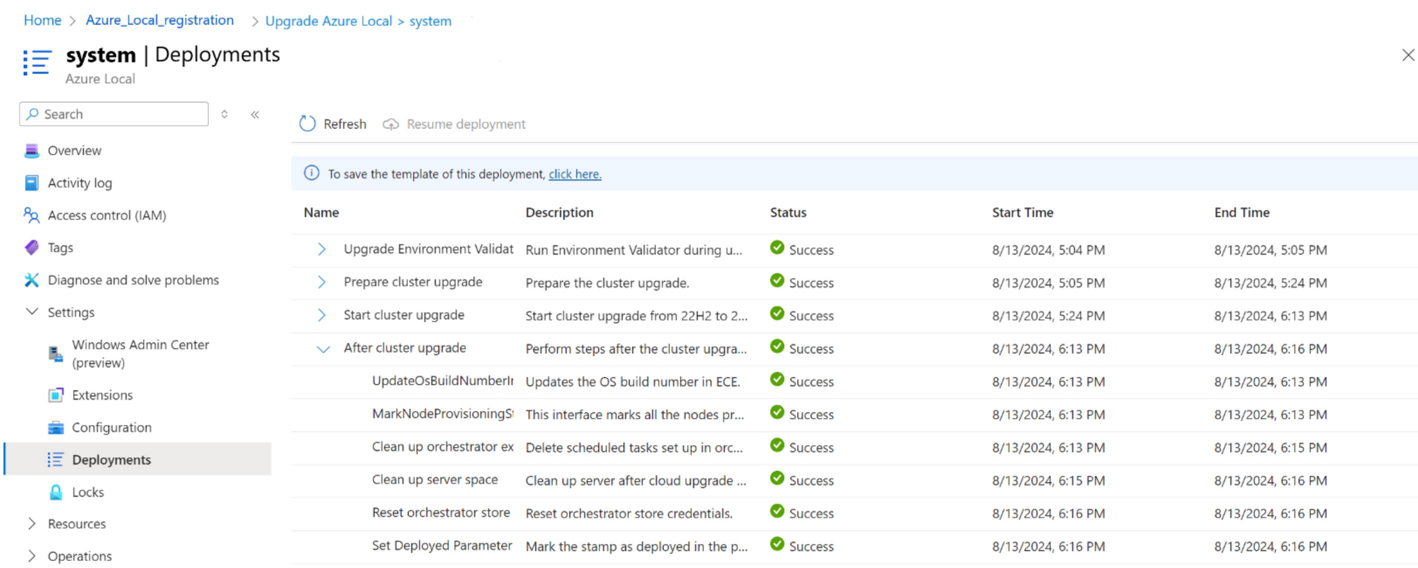


Figure 15. Systems - Deployments

Note

If the upgrade fails, restart the upgrade operation to try again.

3. Verify a successful upgrade.
4. Follow these steps to verify that the upgrade was successful:

   In Microsoft Azure portal, go to the resource group where you deployed the Azure Local instance.

   On the **Overview > Resources** page, you should see the following resources:

Table 3. List of resources to verify after successful upgrade

Resource type	Number of resources
Machine - Azure Arc	1 per machine
Azure Local	1
Arc Resource Bridge	1, -arcbridge suffix by default
Custom location	1, -cl suffix by default
Key Vault	1

Here is a screenshot of the resources in the resource group:

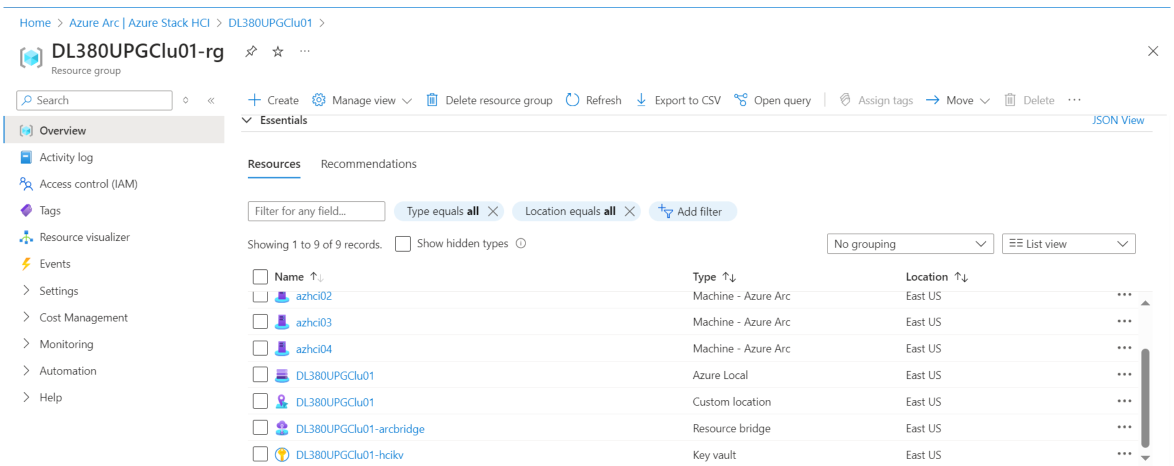


Figure 16. Overview

Step 5.5 Post solution upgrade tasks

Note

As additional services are installed during the solution upgrade, the resource consumption increases after the solution upgrade is complete. After the solution upgrade is complete, additional tasks are required to secure your system and ensure it's ready for workloads.

For Post Solution Upgrade tasks, refer to the **Post Solution Upgrade Tasks section** in the following link Install solution upgrade on Azure Local - Azure Local | Microsoft Learn.

1. To prevent the accidental deletion of resources, you can lock resources. It's recommended that you lock the Arc Resource Bridge.
2. You need to upgrade the security posture. For more information, refer to the following link: Manage security after upgrading your Azure Local from version 22H2 to version 23H2. - Azure Local | Microsoft Learn.
3. You may need to create storage paths for each volume. For more details, refer to the following link : https://learn.microsoft.com/en-us/azure/azure-local/manage/create-storage-path.
4. If you haven't used Cluster-Aware Updating (CAU) for patching your system, you must ensure the permissions are set correctly. For more information, see Cluster aware updating (CAU).

Appendix

1. The QosPolicy and QosTrafficClass name SMBDirect has been renamed to SMB_Direct after installing network ATC.
2. The Jumbopacket value for the storage adapter has been set to its default configuration after enabling network ATC, we have added override to set it to 9014.

   ```
   $StorAdapterPropertyOverrides = New-NetIntentAdapterPropertyOverrides
   $StorAdapterPropertyOverrides.JumboPacket = 9014
   ```
3. After installing the Network ATC, 'NetAdapterQos' has been enabled for all the adapters, we have added override to compute and management intent to disable it.

   ```
   $MgmtAdapterPropertyOverrides = New-NetIntentAdapterPropertyOverrides
   $MgmtAdapterPropertyOverrides.QOS = 0
   ```
4. NetAdapterRdma is enabled for all the adapters after enabling Network ATC, we have added override to compute and management intent to disable it.

   ```
   $MgmtAdapterPropertyOverrides = New-NetIntentAdapterPropertyOverrides
   $MgmtAdapterPropertyOverrides.NetworkDirect = 0
   ```
5. Following the installation of the Network ATC, the 'QosFlowControl' setting for cluster traffic has not been enabled.
6. Modifications have been made to the Validate-DCB tool to update the NIC capability from 10 Gbps to 25 Gbps and to set a bandwidth limit for Live Migration.
7. An override has been added to the storage adapter to prevent the default IP address from being set by the Network ATC.

   ```
   $StorageOverride = New-NetIntentStorageOverrides
   $StorageOverride.EnableAutomaticIPGeneration = $false
   ```
8. Inbox drivers are not supported for use with Azure Local. To determine if your adapter is using an inbox driver, run the following cmdlet. An adapter is considered to be using an inbox driver if the 'DriverProvider' property is set to Microsoft.

Resources and additional links

HPE GreenLake for Microsoft Azure Stack HCI Documentation
hpe.com/support/HPE-GreenLake-MS-AzureStack-HCI
Log in to the HPE Support Center as an HPE Employee to view the internal documents.

HPE Servers
hpe.com/servers

HPE Storage
hpe.com/storage

HPE Networking
hpe.com/networking

HPE GreenLake Advisory and Professional Services
hpe.com/us/en/services/consulting.html

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